

Assignment 8

1.

Write a query that uses a subquery to obtain all orders for the customer named Cisneros. Assume you do not know his customer number (cnum).

```
KD3-99543-Arn timer select * from orders where cnum = (select cnum from customers where cname='Cisneros');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3001 | 18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

7.

Extract all the orders of Motika.

```
KD3-99543-Arn timer select * from orders where snum = (select snum from salespeople where sname='Motika');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
+-----+-----+-----+-----+-----+
1 row in set (0.01 sec)
```

8.

Find all the orders of salespeople servicing customers in London.

```
KD3-99543-Arn timer SELECT *
-> FROM orders
-> WHERE snum IN (
->   SELECT snum
->   FROM customers
->   WHERE city = 'London'
-> );
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
3 rows in set (0.02 sec)
```

6.

Write a query that selects all orders for amounts greater than any for the customers in London.

```
KD3-99543-Arnav select * from orders
-> where amt > ANY (select amt from orders where cnum IN (select cnum FROM customers where city = 'London'));
```

onum	amt	odate	cnum	snum
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3008	4723.00	1990-10-04	2006	1001
3011	9891.88	1990-10-04	2006	1001

6 rows in set (0.03 sec)

4.

Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'. (Hint: find all customers of serres salesperson).

```
KD3-99543-Arnav select * from customers
-> where rating >= ANY(select rating from customers where snum = (select snum from salespeople where sname = 'Serres'));
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.02 sec)

11.

Select customers who have a greater rating than any other customer in Rome.

```
KD3-99543-Arnav select * from customers
-> where rating >= ANY(select rating from customers where snum = (select snum from salespeople where sname = 'Serres'));
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.02 sec)

12.

Select all orders that had amounts that were greater than at least one of the orders from '1990-10-04'.

```

KD3-99543-Arnav select * from orders where amt>ANY(select amt from orders where odate = '1990-10-04');
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 | 309.95 | 1990-10-04 | 2004 | 1002 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)

```

13.

Find all orders with amounts smaller than any amount for a customer in San Jose.

```

KD3-99543-Arnav SELECT * FROM orders
-> WHERE amt < ANY (
->   SELECT amt FROM orders
->   WHERE cnum IN (
->     SELECT cnum FROM customers WHERE city = 'San Jose'
->   )
-> );
+-----+-----+-----+-----+-----+
| onum | amt   | odate   | cnum | snum |
+-----+-----+-----+-----+-----+
| 3001 | 18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 | 767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
| 3009 | 1713.23 | 1990-10-04 | 2002 | 1003 |
| 3007 | 75.75 | 1990-10-04 | 2004 | 1002 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3010 | 309.95 | 1990-10-04 | 2004 | 1002 |
+-----+-----+-----+-----+-----+
8 rows in set (0.00 sec)

```

2.

Write a query to display the order amount, customer names and ratings of all customers where the amount is greater than the average amount from the orders table.

```

KD3-99543-Arnav SELECT o.amt, c.cname, c.rating
-> FROM orders o, customers c
-> WHERE o.cnum = c.cnum
->   AND o.amt > (SELECT AVG(amt) FROM orders);
+-----+-----+-----+
| amt   | cname | rating |
+-----+-----+-----+
| 5160.45 | Liu   | 200   |
| 9891.88 | Clemens | 100   |
| 4723.00 | Clemens | 100   |
+-----+-----+-----+
3 rows in set (0.01 sec)

```

14.

Select those customers whose rating are higher than every customer in London.

```
KD3-99543-Arnab select * from customers where rating > ALL(select rating from customers where city = 'London');
```

cnum	cname	city	rating	snum
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2008	Cisneros	San Jose	300	1007

4 rows in set (0.00 sec)

5.

Write a query that will find all salespeople who have no customers located in their city.
(Hint: Choose the correct keyword ANY/ALL).

```
KD3-99543-Arnab select * from salespeople s
-> where snum not in (select snum from customers where city = s.city);
```

snum	sname	city	comm
1004	Motika	London	0.11
1007	Rifkin	Barcelona	0.15
1003	Axelrod	New York	0.10

3 rows in set (0.01 sec)

9.

Find names and numbers of all salesperson who have more than one customer.

```
KD3-99543-Arnab select sname , snum from salespeople s
-> where (select COUNT(snum) from customers where snum=s.snum)>1;
```

sname	snum
Peel	1001
Serres	1002

2 rows in set (0.00 sec)

10.

Find salespeople number, name and city who have multiple customers. Display the count of customers as well.

```
KD3-99543-Arnab SELECT s.snum, s.sname, s.city, COUNT(c.cnum) AS customer_count
-> FROM salespeople s
-> INNER JOIN customers c ON s.snum = c.snum
-> GROUP BY s.snum, s.sname, s.city
-> HAVING COUNT(c.cnum) > 1;
```

snum	sname	city	customer_count
1001	Peel	London	2
1002	Serres	San Jose	2

2 rows in set (0.02 sec)

3.

Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the amount of the largest order in the table.

```
KD3-99543-Arnav SELECT snum, SUM(amt) AS total_amount
-> FROM orders
-> GROUP BY snum
-> HAVING SUM(amt) > (
->     SELECT MAX(amt)
->     FROM orders
-> );
```

snum	total_amount
1001	15382.07

1 row in set (0.01 sec)